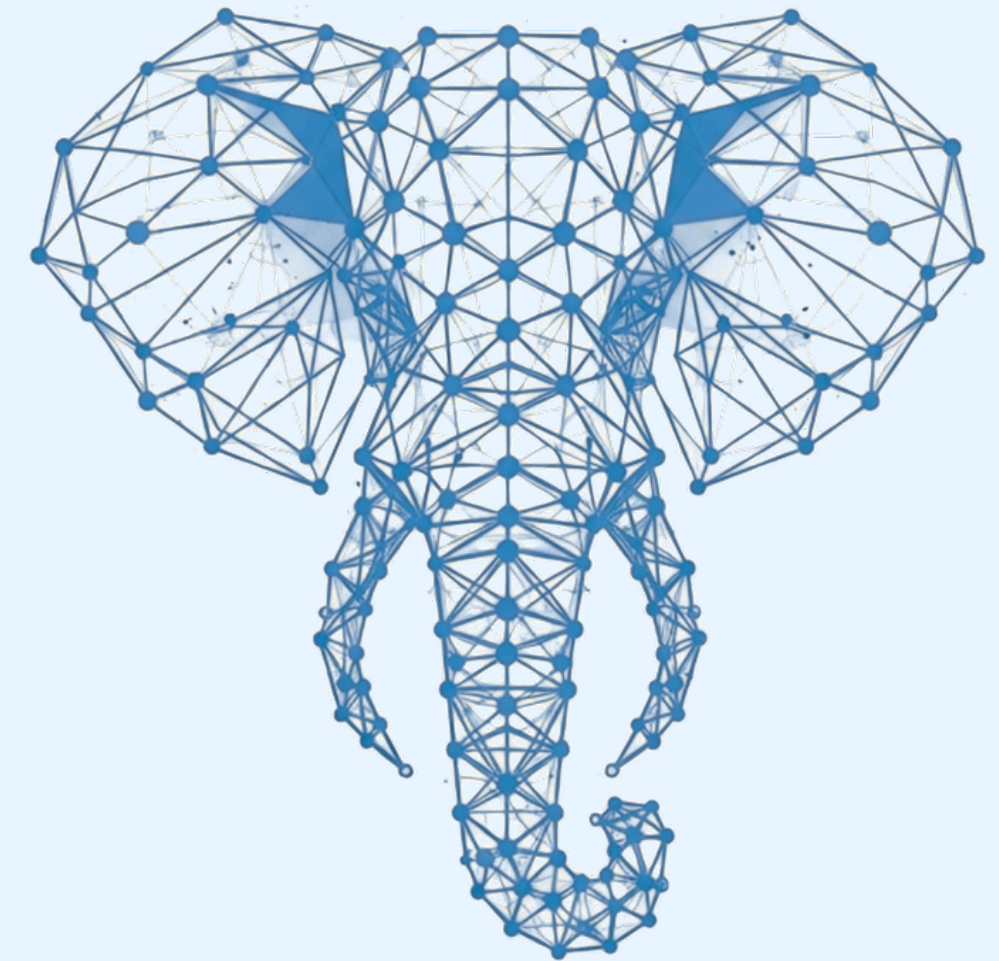


PROJECT 30: UNSUPERVISED DOMAIN ADAPTATION FOR IMAGE RECOGNITION UNDER DOMAIN SHIFT

Gianluca Diquattro

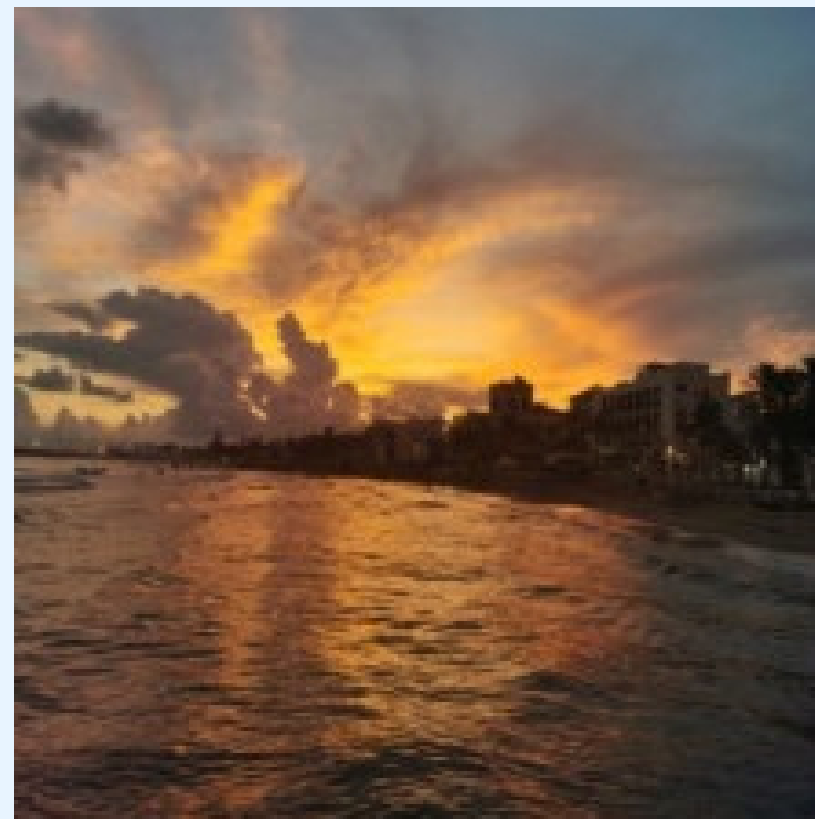
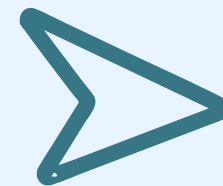
<https://github.com/TonyEffee/progetto-uda-deeplearning>



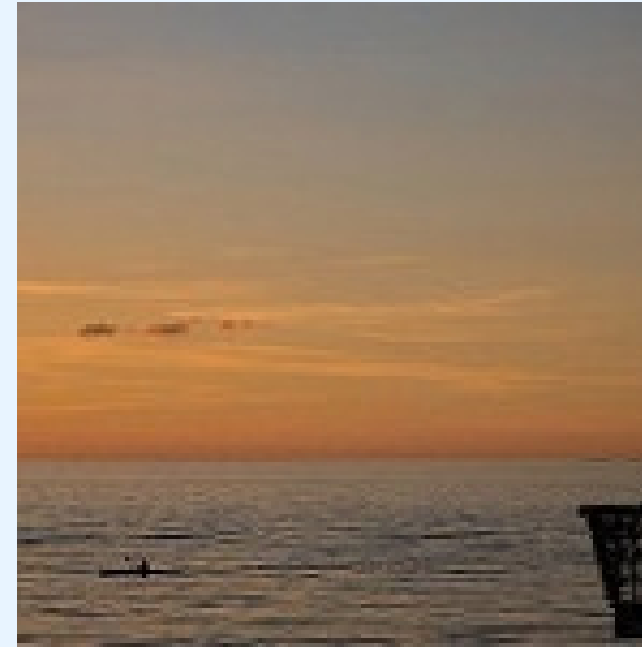
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ADVANCED MODELS AND METHODS

**DEEP LEARNING: ADVANCED
MODELS AND METHODS (DL26)**

Prof. A. Furnari

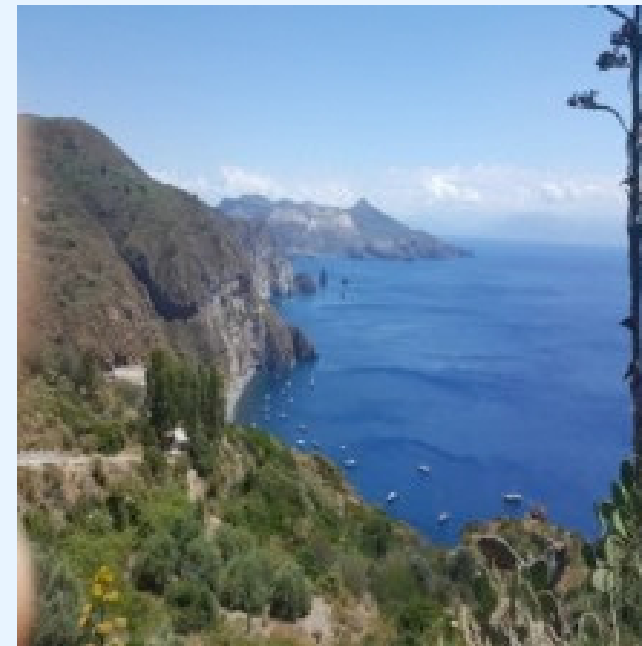


IL MIO MODELLO FUNZIONA AL 93%, FINCHÉ NON GLI MOSTRO LE MIE FOTO



92.7%

sulle immagini
del dataset



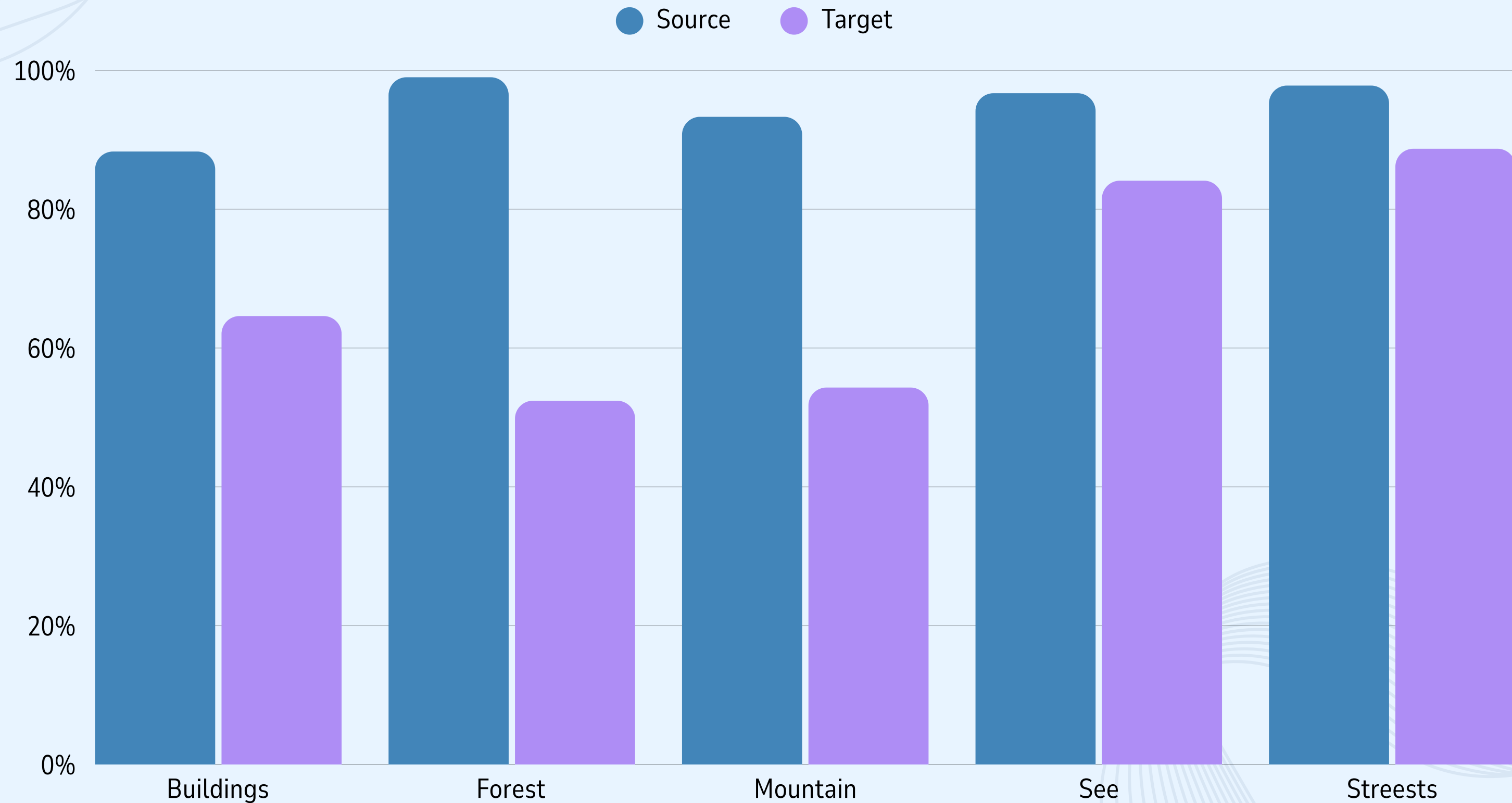
74.1%

sulle mie immagini

DATI: DUE DOMINI, CINQUE CLASSI IN COMUNE

SOURCE: INTEL IMAGE CLASSIFICATION	TARGET: DATASET PERSONALE
14.034 immagini di training	1.210 immagini
3.000 di validazione	nessuna etichetta usata in training
6 classi, bilanciate	5 classi, sbilanciate
150×150 px	224x224 px

IL DOMAIN SHIFT NON COLPISCE TUTTE LE CLASSI ALLO STESSO MODO



QUATTRO STRATEGIE, NESSUNA ETICHETTA DEL TARGET

01

CORAL

02

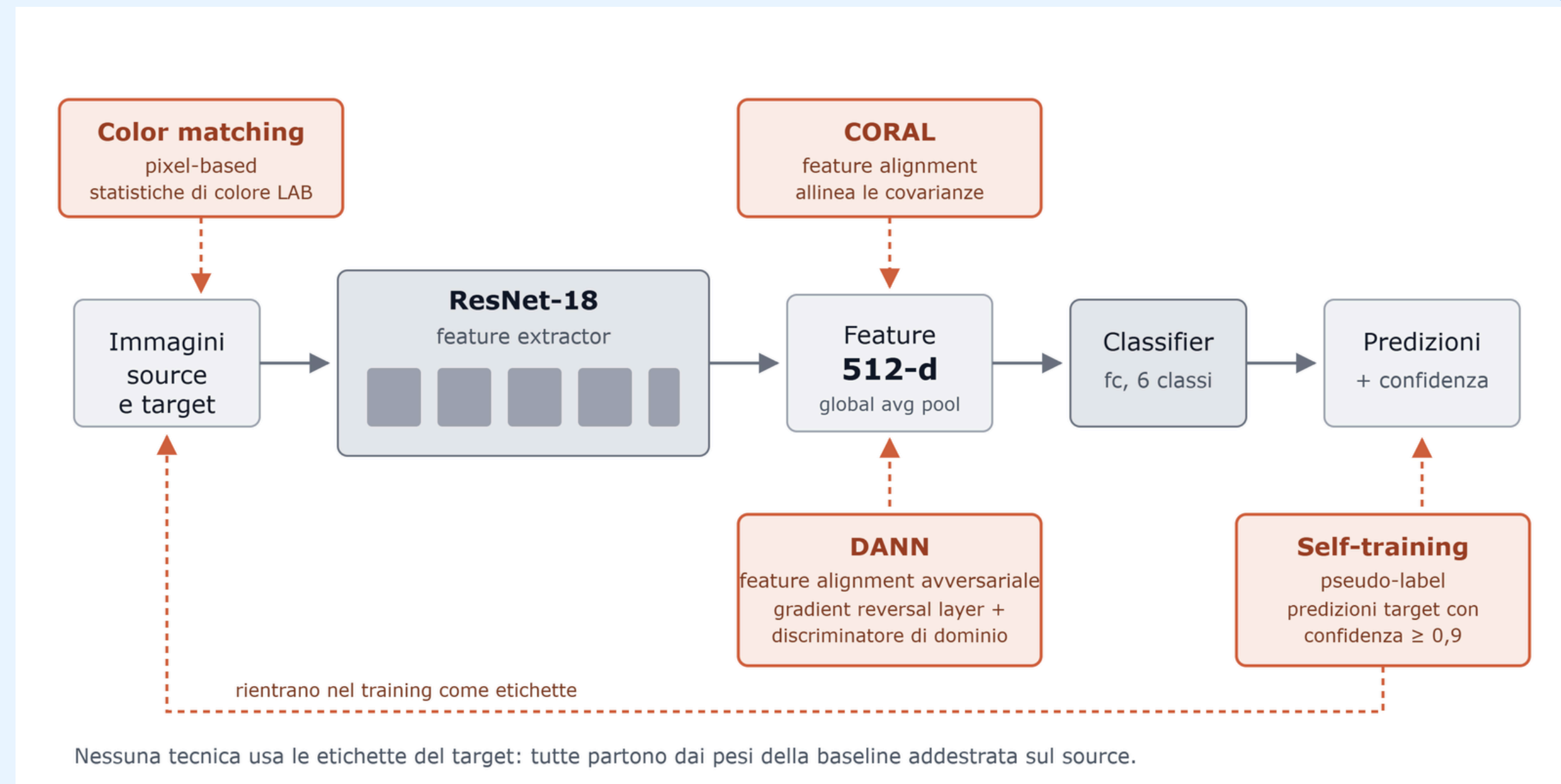
DANN

03

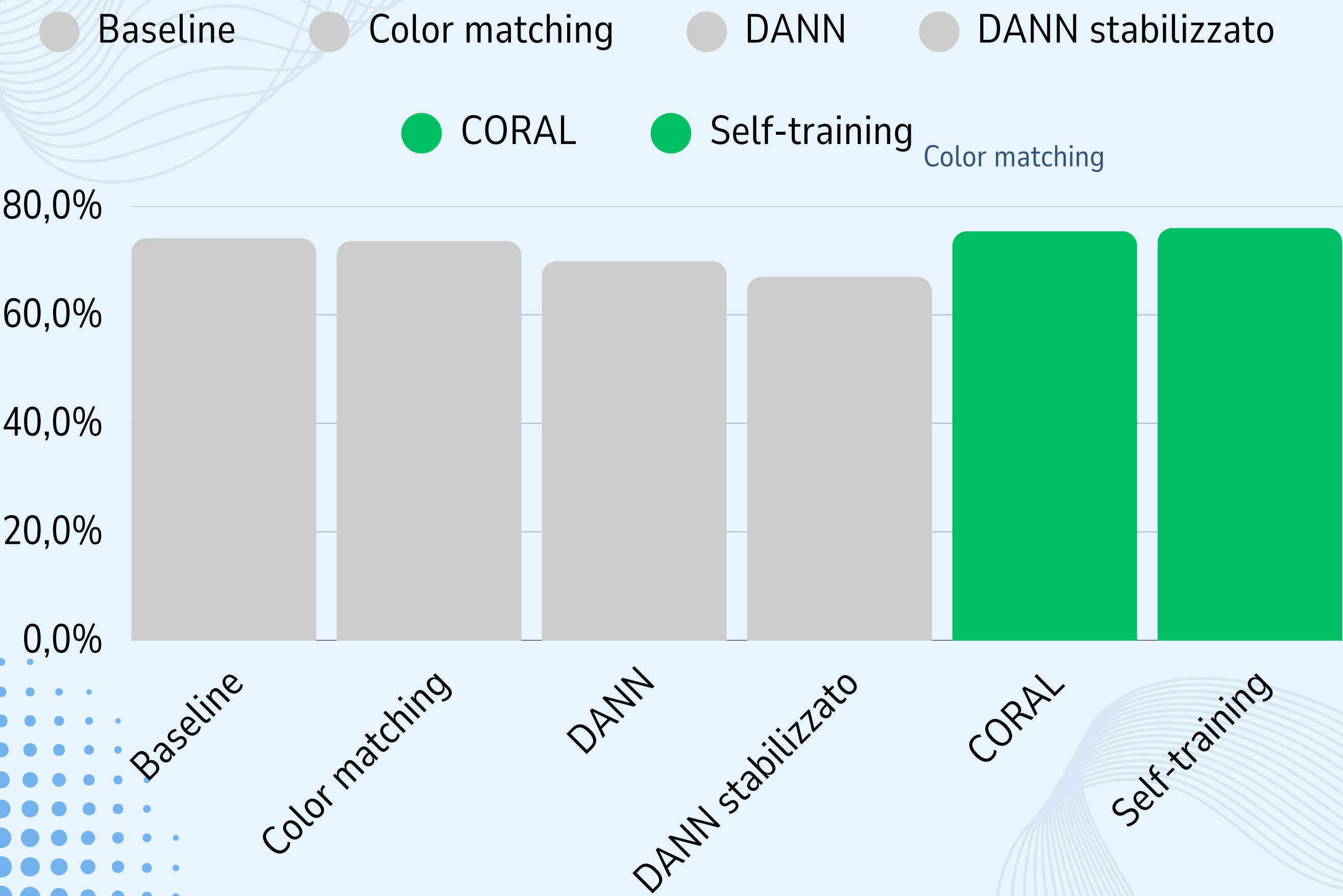
SELF-TRAINING

04

COLOR MATCHING

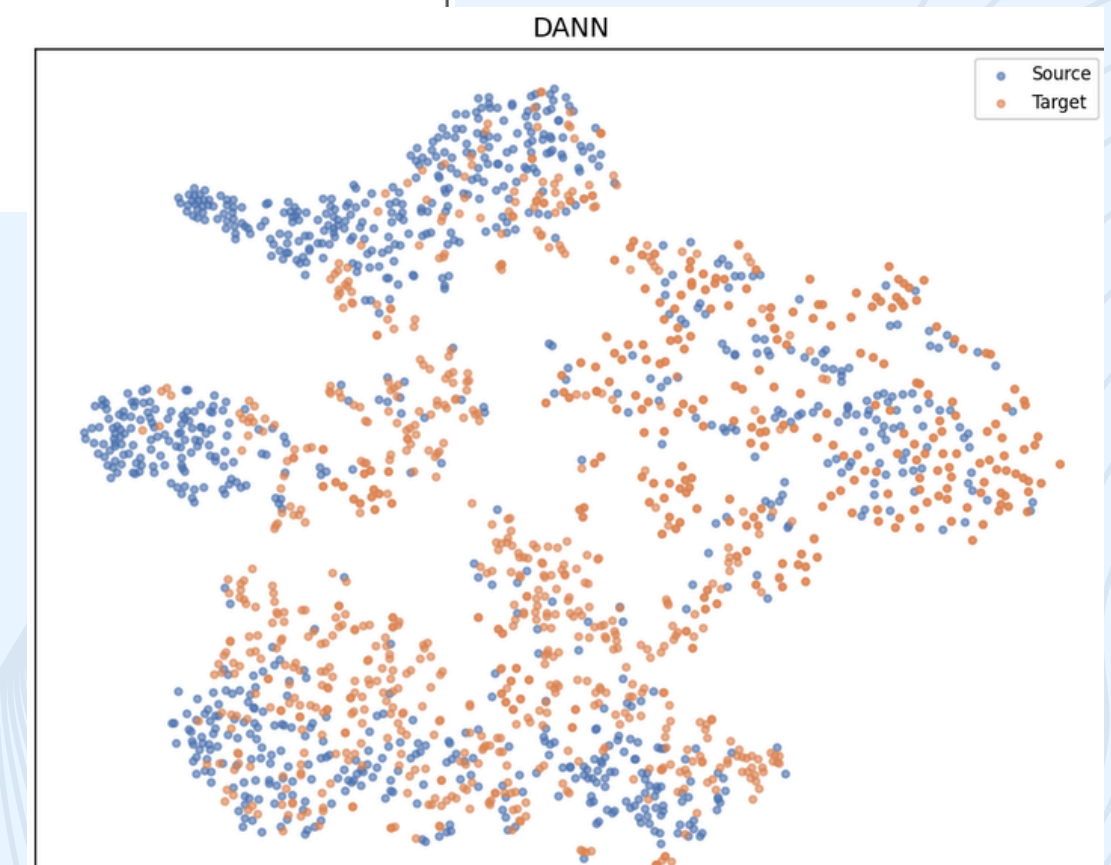
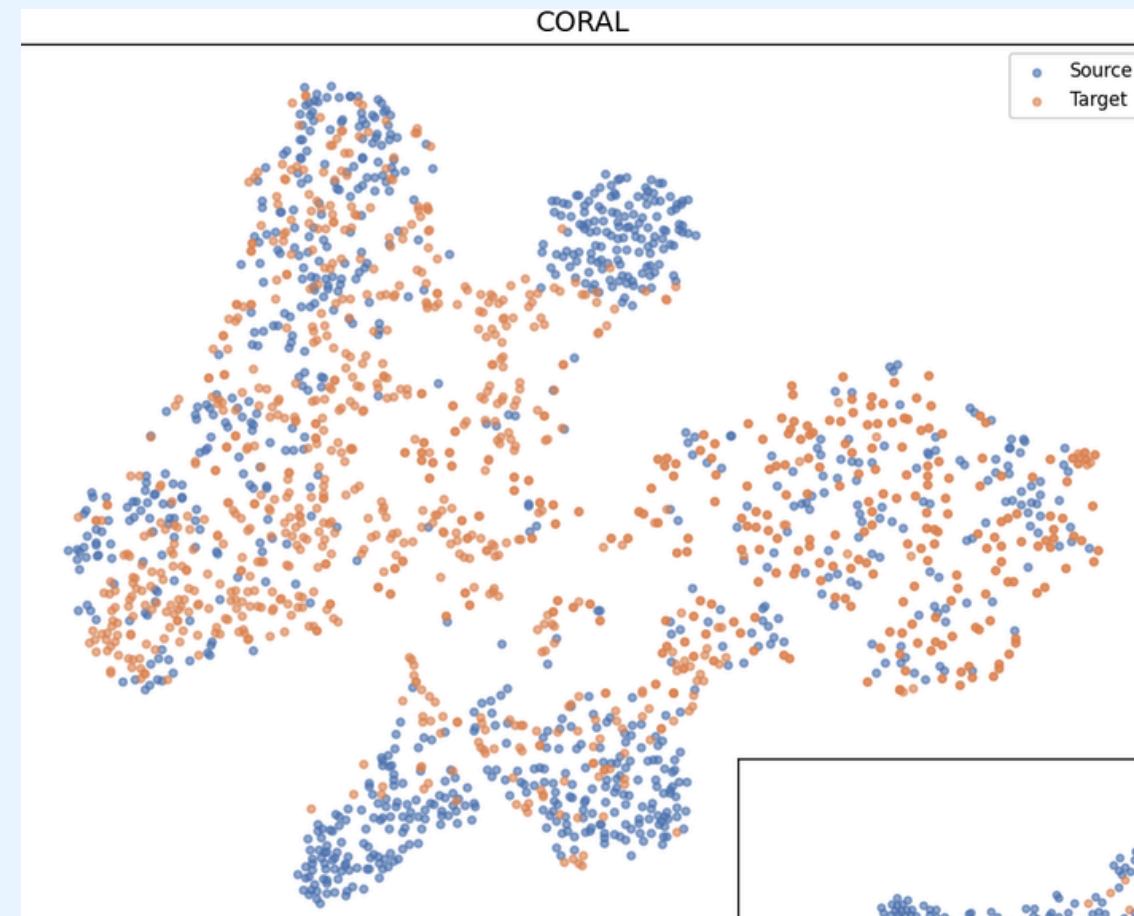
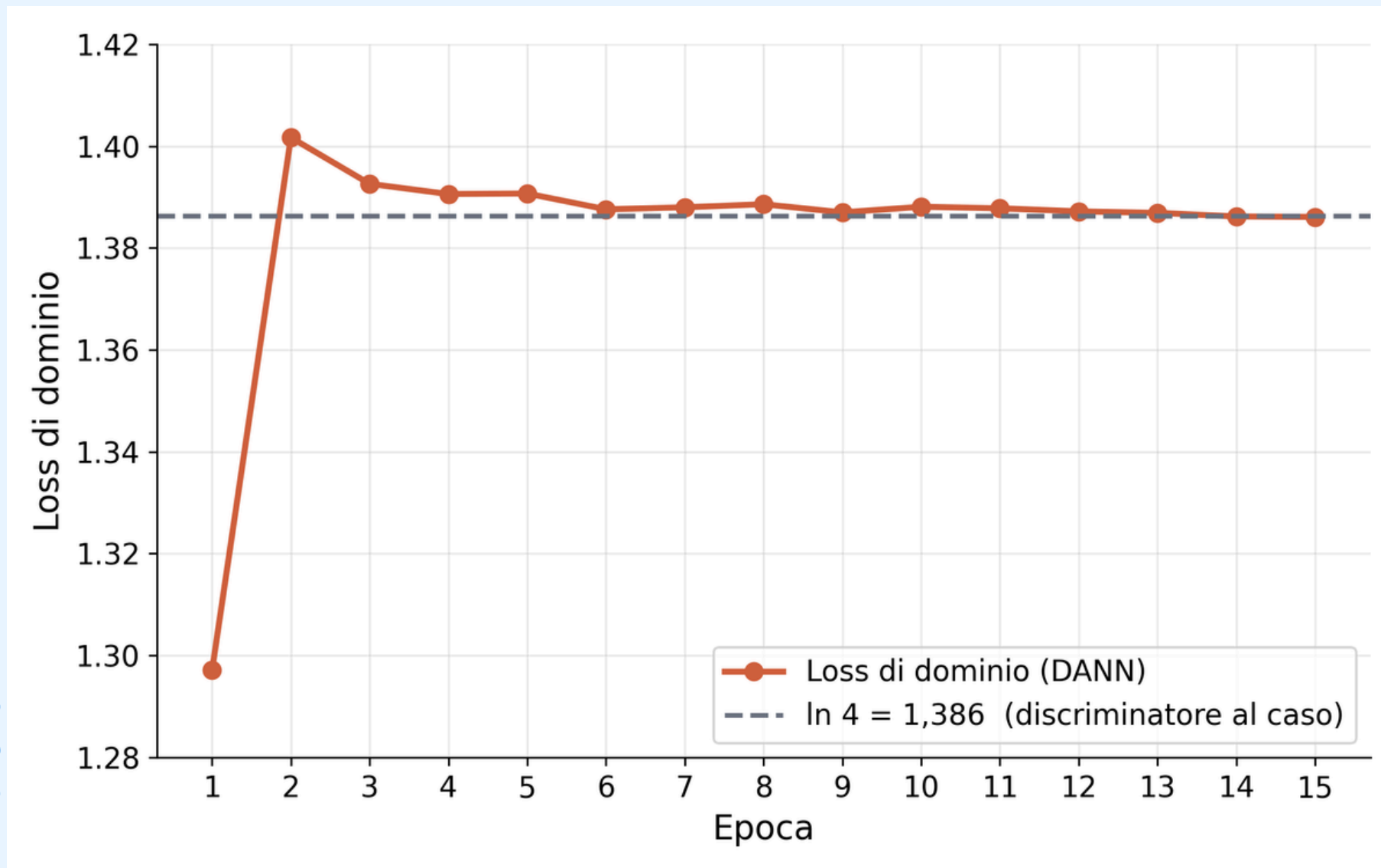


ALLINEAMENTO STATISTICO MEGLIO DI TUTTI



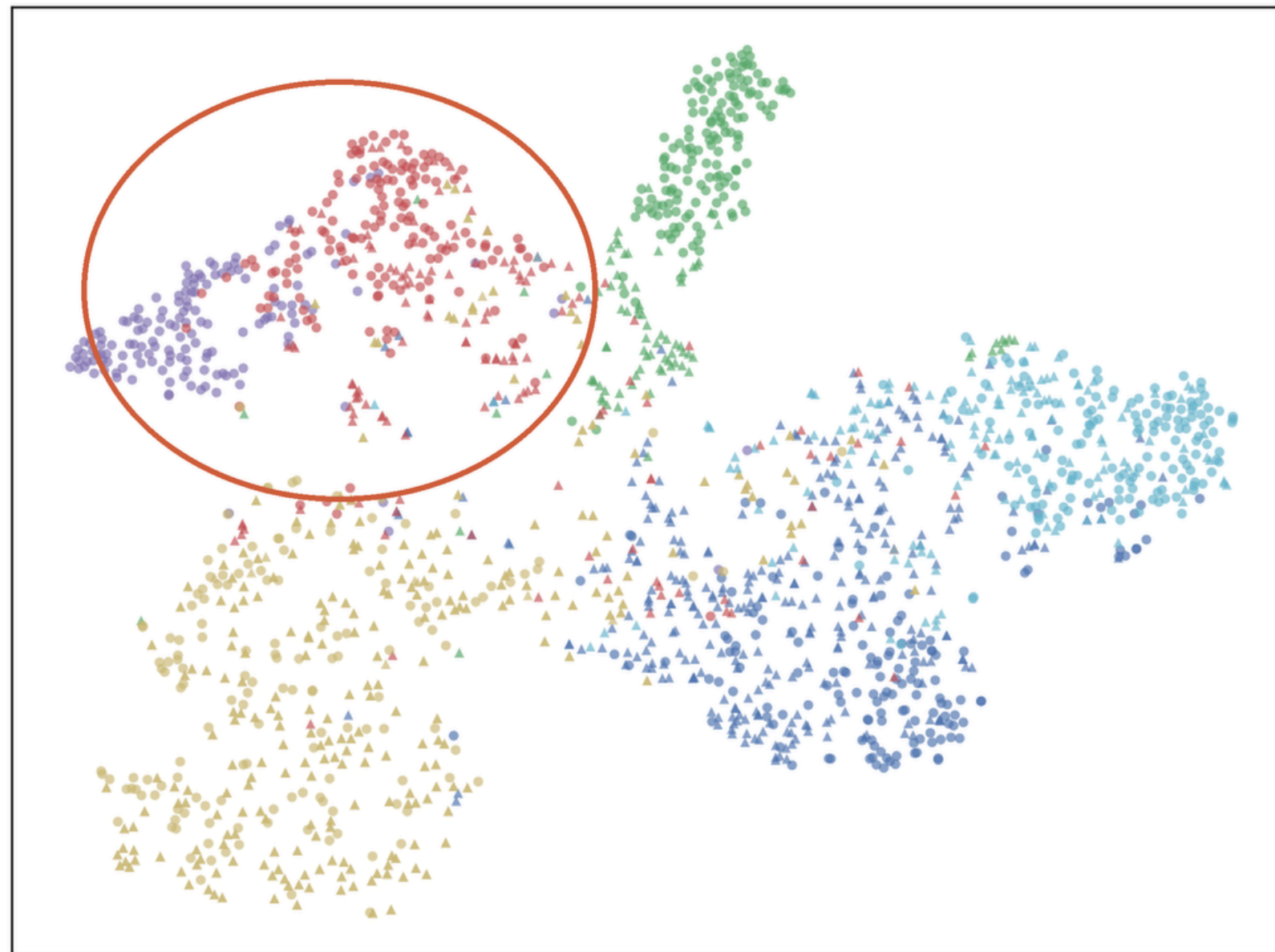
MODELLO	Δ
COLOR MATCHING	-0.4
DANN	-4.1
DANN STABILIZZATO	-7.0
CORAL	+1.3
SELF TRAINING	+1.9

DANN INGANNA IL DISCRIMINATORE, MA NON SERVE A CLASSIFICARE MEGLIO



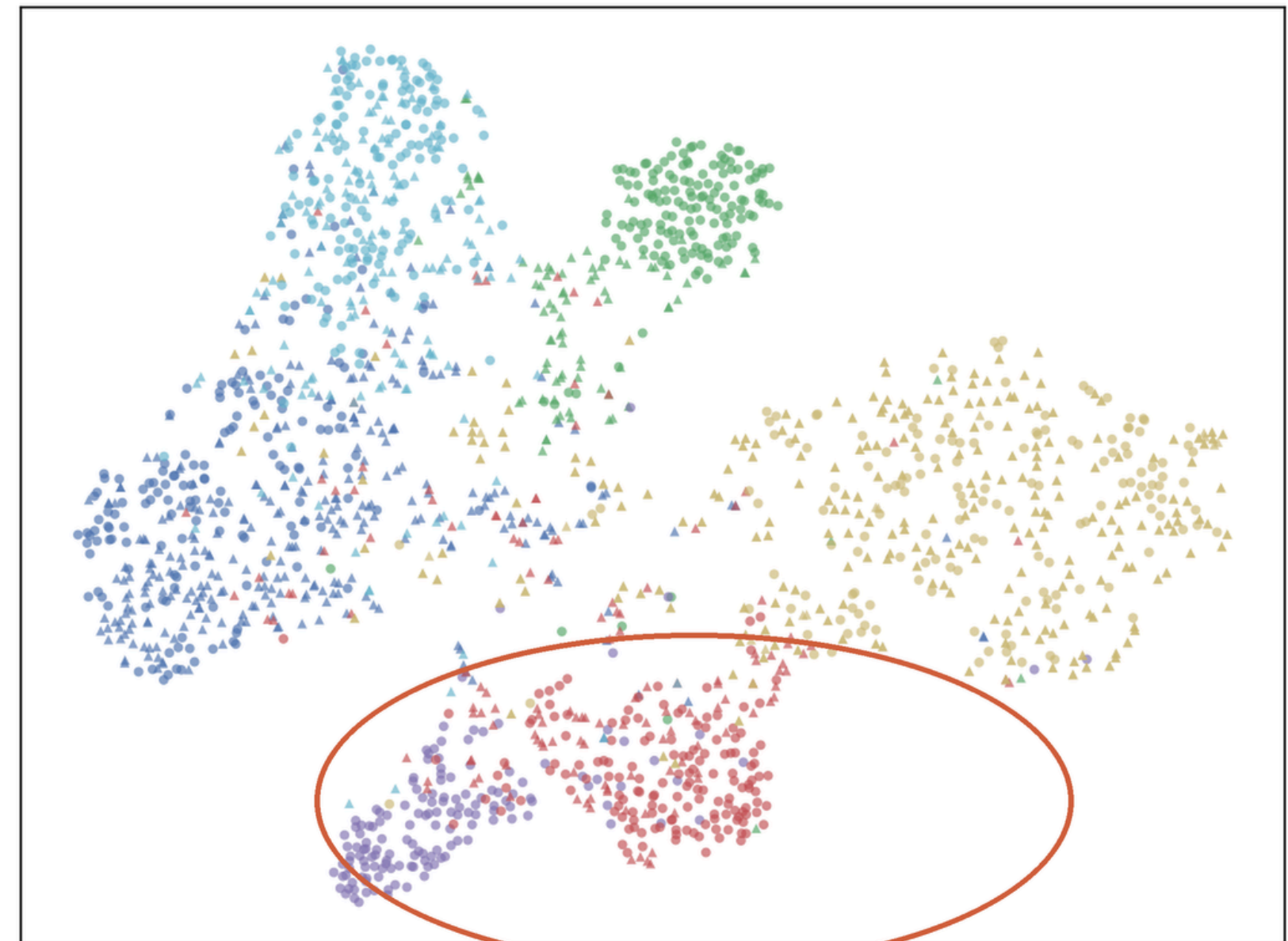
PARTE DEGLI ERRORI NON DIPENDE DAL DOMAIN SHIFT

Baseline



● buildings ● forest ● glacier ● mountain ● sea ● street

CORAL



● source ▲ target

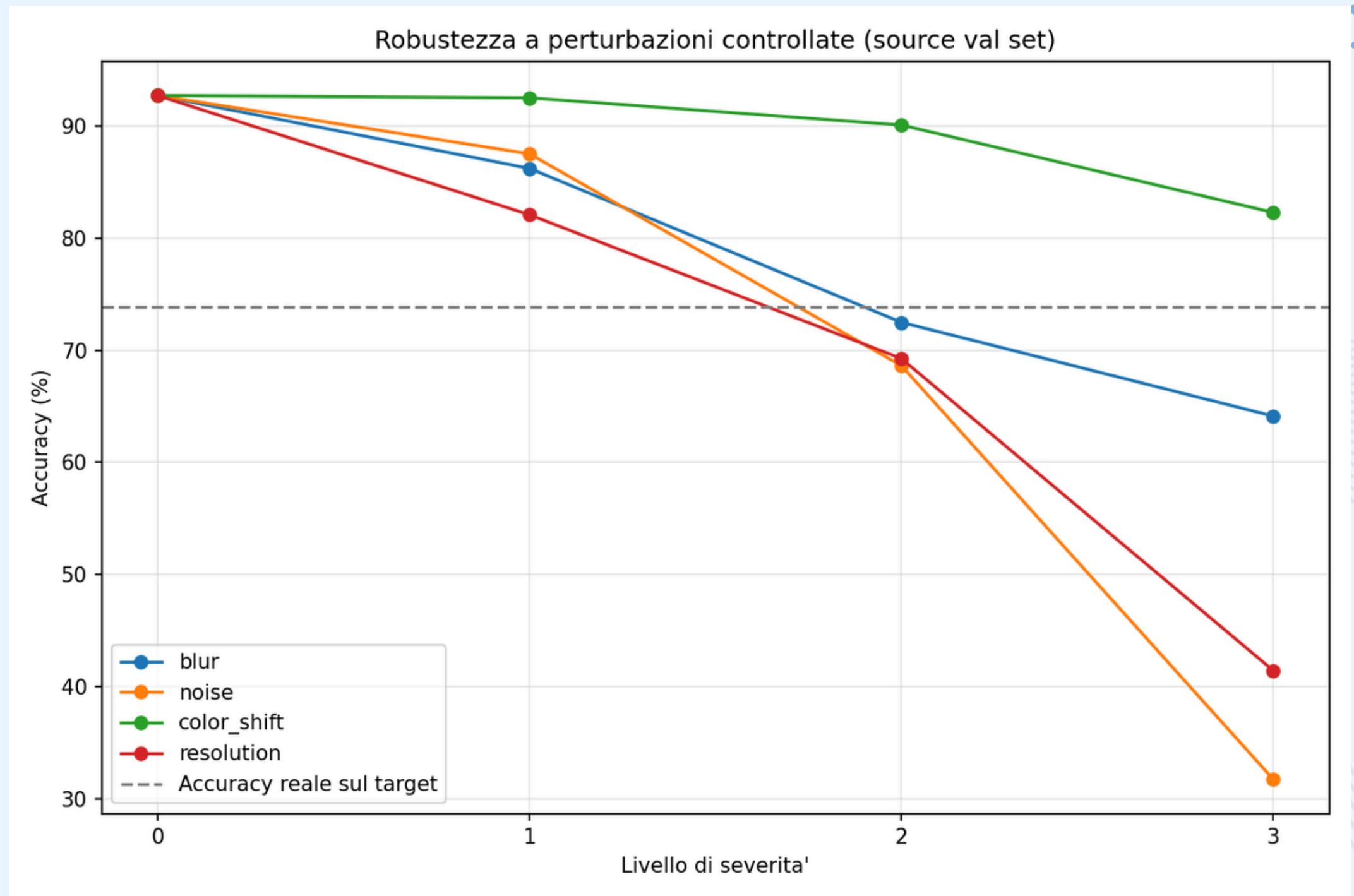
LO SHIFT È QUESTIONE DI DETTAGLIO, NON DI COLORE

01 COLOR-SHIFT

02 NOISE

03 BLUR

04 RESOLUTION



CONCLUSIONI

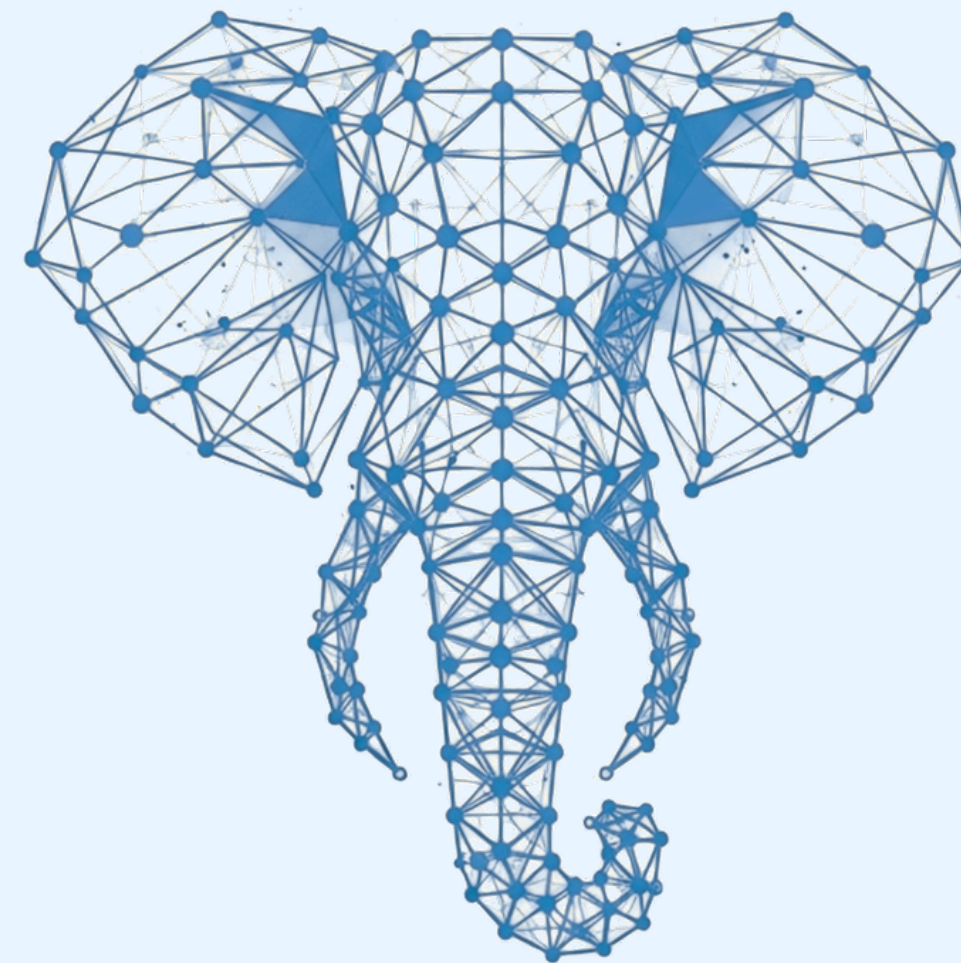
Su un domain shift reale ho recuperato 2 punti su 19

takeaway:

- L'allineamento statistico batte quello adversarial quando il target è piccolo e sbilanciato
- Misurare quali fattori causano lo shift indirizza le scelte: qui servirebbero augmentation su sfocatura e risoluzione, non sul colore.
- Alcuni errori vanno corretti nel source, prima di provare ad adattare

Limiti e lavoro futuro: un solo seed, setting trasduttivo, target piccolo; in futuro tecniche condizionate alla classe (CDAN), augmentation mirate, più foto personali per le classi meno rappresentate

GRAZIE PER L'ATTENZIONE



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